Fan Xiao

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Education

Ludwig-Maximilians-Universität München, Munich, Germany 2023.09 – 2027.03 (expected)
PhD candidate in Medical Physics

Research focus: Online Adaptive Radiotherapy, AI-accelerated Dose Calculation, Photon/Proton Monte Carlo Simulation

Southern Medical University, Guangzhou, China 2020.09 – 2023.06
MS in Biomedical Engineering, GPA 3.6/4.0

Wuhan University of Technology, Wuhan, China 2016.09 – 2020.06
BS in Communication Engineering

Research Experience

DoseRAD2026 MICCAI Challenge | Organizer 2025.09 – Now

- Built the first open-source beam-level photon/proton MC dose dataset (paired CT/MRI, abdomen & thorax) for the MICCAI 2026 DoseRAD Challenge
- Pipeline: MR-CT deformable registration, air-cavity correction, matRad planning; Geant4 ground truth for 40,500 photon beams and 81,000 proton beamlets
- Published in ***Medical Physics*** **(2026)**, DOI: 10.1002/mp.70714

Multi-model study of fast VMAT segment dose calculation | First Author 2025.07 – 2026.04

- Benchmarked Beam's Eye View (BEV) vs. patient-coordinate DL dose calculation across five models on a Geant4 segment dose dataset (24 prostate VMAT plans)
- BEV methods were markedly more robust; two lightweight models (CNN-Mamba, CNN-ConvLSTM) take 39–51 ms per segment and 5–6 s per full plan, with >99% local gamma (1%/3 mm)
- Published in ***Physics in Medicine & Biology*** **(2026)**, DOI: 10.1088/1361-6560/ae6413

DL-based synthetic-CT-free photon dose calculation: proof-of-concept | First Author 2024.10 – 2025.06

- Proposed a UNet-LSTM framework that predicts photon beamlet dose directly from MRI, avoiding synthetic CT generation (34 prostate patients, 0.35 T MR-Linac)
- 12 ms per beamlet; 2%/2 mm gamma passing rate 99.68% for beamlets and >99.80% for plans
- Published in ***Medical Physics*** **(2025)**, DOI: 10.1002/mp.70106

Proton Range Verification via LSTM-based Prompt Gamma Prediction | First Author 2024.01 – 2024.09

- Developed an LSTM model using stopping power maps and proton dose to predict prompt gamma emissions, replacing MC simulation (20–30 CPU-hours per pencil beam)
- Range shift error 0.15 mm, gamma 98.5% (2%/2 mm), <130 ms per pencil beam over 125–210 MeV
- Published in ***Physics in Medicine & Biology*** **(2024)**, DOI: 10.1088/1361-6560/ad8e2a

LSTM-based proton dose calculation in a magnetic field | Co-First Author 2023.10 – 2024.06

- Developed a multi-energy LSTM proton dose engine for MR-guided proton therapy in a 1.5 T field
- Gamma (2%/2 mm) 99.7–99.9% for beamlets and 97.9–99.6% for plans; 9–10 ms per pencil beam
- Published in ***Physics in Medicine & Biology*** **(2024)**, DOI: 10.1088/1361-6560/ad7f1e

Technical Skills

- Programming languages: Python, MATLAB, C++ (Geant4), Bash
- Deep Learning: PyTorch, LSTM/ConvLSTM, Transformer, Mamba, U-Net/nnU-Net
- Tools: Git, Docker, Slurm, CuPy, Plastimatch, pydicom
- Treatment planning: RayStation (scripting), matRad (VMAT/IMPT scripting), Varian Eclipse

Additional Info and Awards

- Conference experience:
  MICCAI 2026 Oral (Strasbourg), ESTRO 2026 Poster (Stockholm),
  AAPM 2025 Oral (Washington, DC), ICCR 2024 Oral (Lyon), ECMP 2024 Oral (Munich)
- Awards:
  Chinese National Scholarship (2022) – Top 2%
  First-class Academic Scholarship (2022, 2023) – Top 20%
- Languages:
  Chinese (Native), English (Working language)